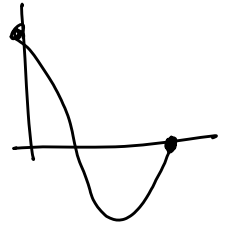
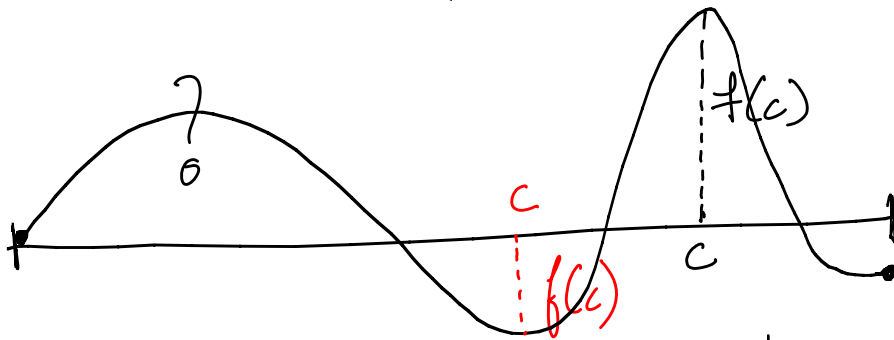


Unit 4.1 Extreme values

Note Title

2015/04/15

Absolute maximum / Absolute minimum

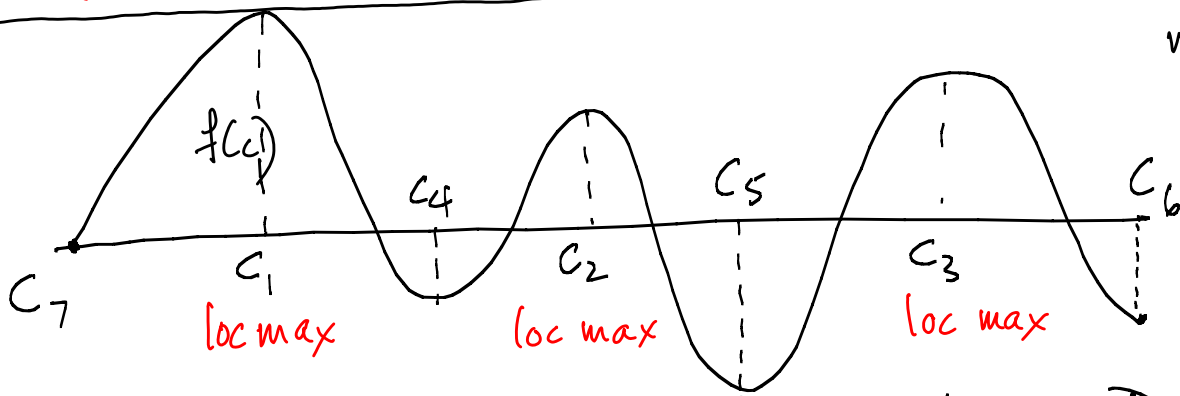


Let D be the domain of the function f .

- $f(c)$ is the absolute maximum of f on D if $f(c) \geq f(x)$ for all $x \in D$.
- $f(c)$ is the absolute minimum of f on D if $f(c) \leq f(x)$ for all $x \in D$.

Local maximum & local minimum

(Also called relative max/min)



$f(c)$ is a local maximum of f on D if $f(c) \geq f(x)$ for all x values close to c .

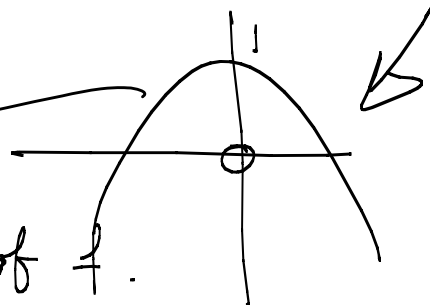
$f(c)$ is a local minimum of f on D if $f(c) \leq f(x)$ for all x values close to c .

We say and write:

- An absolute / local maximum / minimum is called an **extreme**
eg. Find the local extremes.....
local max / min.

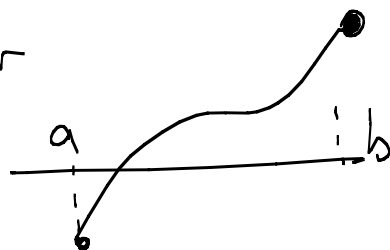
- An abs/local extreme is a **function value**

$$f(x) = -x^2 + 1$$



- NB. $f(0) = 1$ is the absolute max of f .
or f has an abs max of 1 at $x=0$.

- An absolute max/min can occur at the **endpoints** of the interval
Abs max: $f(b)$ abs min: $f(a)$



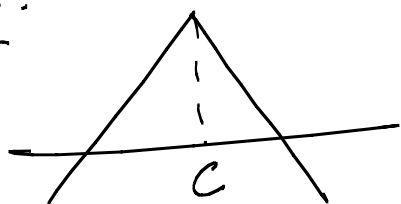
- An extreme can occur in more than one place

NB Critical numbers — x -values.

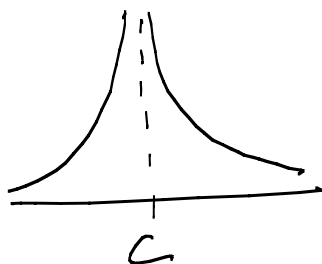
A critical number of a function f is a number c in the domain of f such that $f'(c) = 0$ or $f'(c)$ does not exist.

NB: c is in D

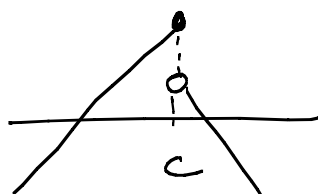
Eg.



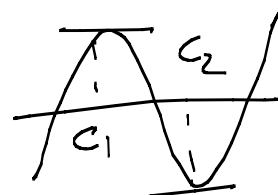
c is a crit. number



No crit numbers.



c is a crit number.



2 crit numbers.

Eg. Find the critical numbers and decide whether f has an extreme at the point or not.

1. $f(x) = x^3 - 3x$

$$\Rightarrow f'(x) = 3x^2 - 3$$

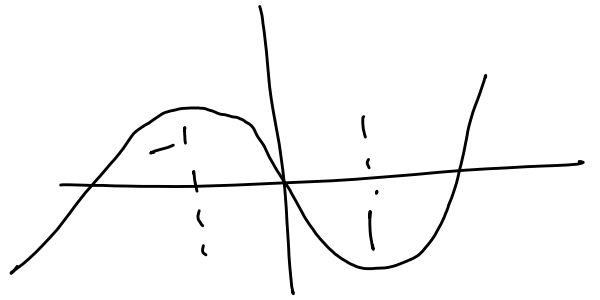
$$f'(x) = 0$$

$$\Rightarrow 3(x^2 - 1) = 0$$

$$\Rightarrow x = \pm 1$$

$\Rightarrow f$ has a local max in $x = -1$, namely 2.

f has a local min in $x = 1$, namely -2.

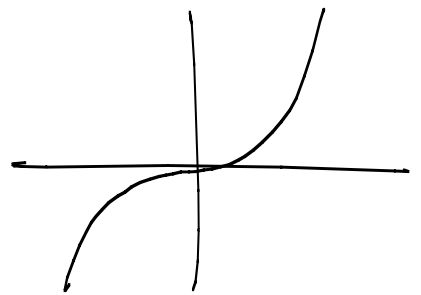


2. $f(x) = x^3$

$$f'(x) = 3x^2 = 0$$

Crit number: $x = 0$

f does not have an extreme in $x = 0$



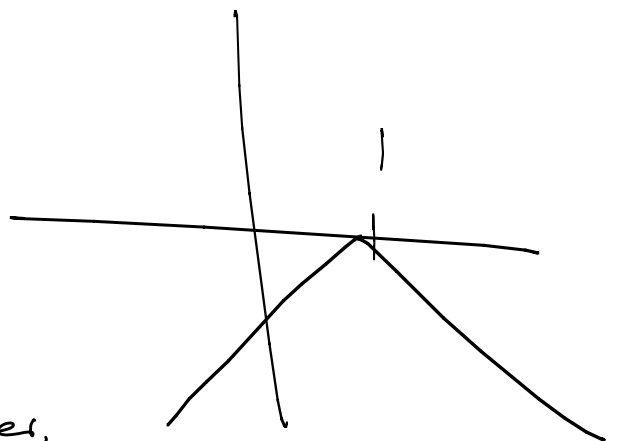
3. $f(x) = -|x-1|$

f' does not exist
in $x = 1$.

f is defined in $x = 1$

So $x = 1$ is a crit number.

$f(1)$ is a local max and also an abs max.



How can you be certain that the function has absolute extremes?

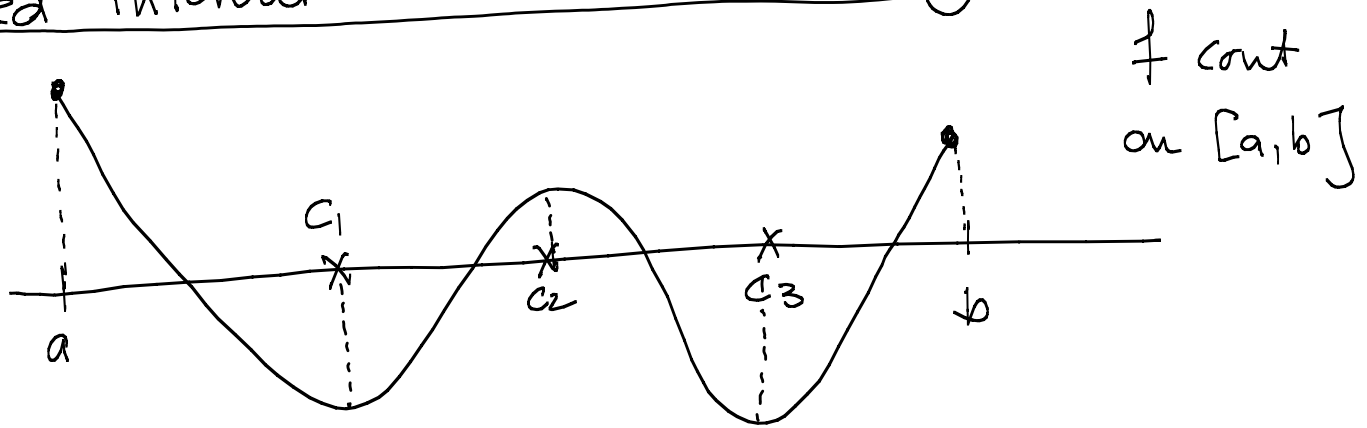
The following theorem answers the question

Extreme value theorem (Fermat's theorem)

If f is continuous on a closed interval then f has an absolute maximum and an absolute minimum.



Closed interval method for finding extremes:



- AB
1. Check that f is continuous on $[a, b]$
 2. Find the critical numbers of f on $[a, b]$
 3. Find the function values at the critical numbers and also find $f(a)$ & $f(b)$
 4. The smallest number in 3. is the absolute minimum and the biggest number in 3 is the absolute maximum.

Examples: Use the Closed Interval Method to find the absolute extremes of

① $f(x) = x^4 - 2x^2$ on $[-\frac{1}{2}, 2]$

1. f cont on the interval (polynomial)

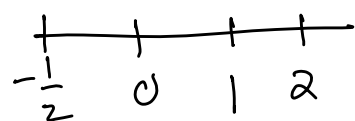
2. Crit numbers:

$$f'(x) = 4x^3 - 4x = 0$$

$$\Rightarrow 4x(x^2 - 1) = 0$$

$$\Rightarrow \underline{x=0} \text{ or } \underline{x=1} \text{ or } x \neq -1$$

Not in the interval.



3. $f(-\frac{1}{2}) = -\frac{7}{16}$

$$f(0) = 0$$

$$f(1) = -1$$

$$f(2) = 8$$

4. f has an absolute maximum $f(2) = 8$

f has an absolute minimum $f(1) = -1$.

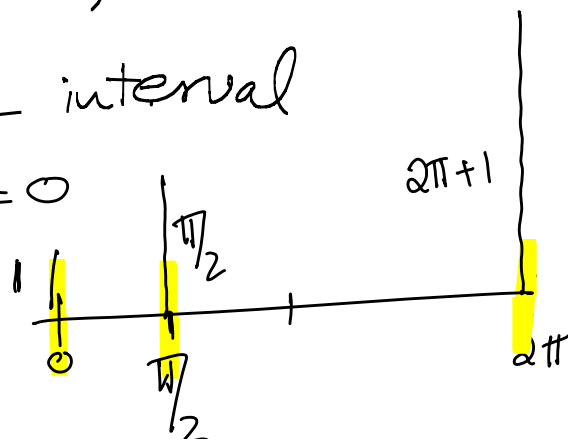
2. $f(x) = x + \cos x$ on $[0, 2\pi]$.

1. f is cont on the interval

2. $f'(x) = 1 - \sin x = 0$

$$\Rightarrow \sin x = 1$$

$$\Rightarrow x = \frac{\pi}{2}$$



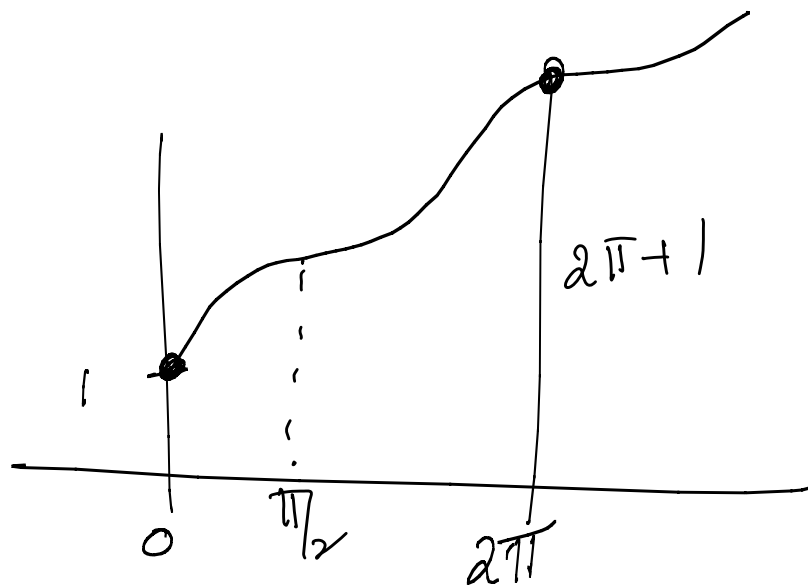
3. $f(0) = 1$

$$f(\pi/2) = \pi/2$$

$$f(2\pi) = 2\pi + \cos 2\pi = 2\pi + 1 \simeq 7$$

4. f has an absolute max $f(2\pi) = 2\pi + 1$

f has an absolute min $f(0) = 1$.



Recapitulating: (f is defined on domain D)

f has an absolute maximum in c if
 $f(c) \geq f(x)$ for all $x \in D$

f has a local maximum in c if
 $f(c) \geq f(x)$ for all x close to c .

f has a critical point c if c is in the domain of f and $f'(c) = 0$ or $f'(c)$ is not defined.

Horizontal line:

